From: **Y Combinator** apply@ycombinator.com
Subject: YC Summer 2021 Application Submitted
Date: March 20, 2021 at 1:19 AM
To: ivan@ambientsystems.ai

Hello **engID**,

Your Y Combinator Summer 2021 application for Ambient Systems has been submitted for review.

We've included the text of your application below.

Thanks,
YC

https://www.ycombinator.com/apply

COMPANY

Company name:

Ambient Systems

Company url, if any:

https://www.ambientsystems.ai/

If you have a demo, what's the url? Demo can be anything that shows us how the product works. Usually that's a video or screen recording.

(Please don't password protect it; just use an obscure url.)

https://youtu.be/03AIMYF-wn0

Describe what your company does in 50 characters or less.

Building automation technology for indoor farms

What is your company going to make? Please describe your product and what it does or will do.

Ambient is making an end-to-end solution for climate management of indoor farms. With increasing pressures of climate change and land constraints in urban centers, indoor farms are becoming a viable option for locally grown food. However, there are many challenges when it comes to running an indoor farm. In particular, there is a large gap in how indoor farm operators manage their air conditioning and mechanical ventilation (ACMV). These challenges are different from the ones faced by commercial and residential buildings. Modern Building Management Systems (BMS) and Building Automation Systems (BAS) need to catch-up with this growing global trend. Ambient provides a streamlined way for customers to optimize the indoor climate through its proprietary solution enabled IoT and .

Where do you live now, and where would the company be based after YC?

(List as City A, Country A / City B, Country B.)

Singapore / Singapore

CONTACT

Personal email address of the founder who is filling out this application:

Please enter an email address that you check often and that you will have access to for a long time.

ivan@ambientsystems.ai

Phone number(s):

65[redacted]

FOUNDERS

Founders

HACKER NEWS USERNAME	NAME	EMAIL	PROFILE UPDATED
	Ivan		

engID	Ivan Damnjanovic	ivan@ambientsystems.ai	yes

Please enter the url of a 1 minute unlisted (not private) YouTube video introducing the founder(s). This video is an important part of the application. (Follow the Video Guidelines.)

https://www.youtube.com/watch?v=uxFrW5GI2Yo

How many founders are on the team?

(Fill out this number of founder profiles)

1

CATEGORY

Which category best applies to your company?

Other

PROGRESS

How far along are you?

We've completed market validation, assembled the team, prepared patent filing with IP lawyers, and begun MVP pilot with an indoor farm operator in Singapore

How long have each of you been working on this? How much of that has been full-time? Please explain.

I have been working on it for about a year (4 months full-time and 8 months part-time). Some of the advisers have been there for that duration. Few more part-time team members joined in the last two months.

Are people using your product?

Yes

How many active users or customers do you have? If you have some particularly valuable customers, who are they? If you're building

particularly valuable customers, who are they? If you're building hardware, how many units have you shipped?

We have LOIs with two indoor farm operators in Singapore. We are collecting data with the customer at this time. This pilot will be used to test and calibrate the ML-enabled optimization models.

Several other types of building operators have also provided validation for this idea (i.e., confirmed that there is a need). Namely, one of the largest healthcare facility managers in Southeast Asia has similar challenges in terms of achieving optimal indoor climate efficiently at low cost.

Do you have revenue?

No

Anything else you would like us to know regarding your revenue or growth rate?

We plan to grow with the customer. As their operations expand, we plan to add more features to the initial product. Ambient's business model consists of CAPEX and OPEX components. Customers pay for the related hardware necessary to enable the system up-front, as a capital cost. Then the software platform and optimization model is delivered through a subscription model. The pricing for the product will be done on a per-area basis, meaning that larger facilities pay more for the service than smaller ones. We are currently refining the exact numbers based on customer conversations, but we can discuss further, along with financial projections, in the follow-on conversations.

If you are applying with the same idea as last batch, did anything change? If you applied with a different idea, why did you pivot and what did you learn from the last idea?

I initially applied with an idea for 'Ambient Smart Homes', but pivoted towards the industrial sector due to a greater need for a climate management solution. I extensively tested the B2B idea for healthcare and industrial sectors, namely indoor farms.

If you have already participated or committed to participate in an incubator, "accelerator" or "pre-accelerator" program, please tell us

about it.

At the moment we participating at a university incubator for deep-tech startups. We plan to pitch for a pre-seed round of $150k in May.

IDEA

Why did you pick this idea to work on? Do you have domain expertise in this area? How do you know people need what you're making?

When I started Ambient a year ago, the focus was on enhanced indoor air quality for residential buildings (i.e., a smart home product). My background in building science and software was suited to solve the gap in the market using advanced technology. However, as I spoke with more customers, I realized that the need for better climate management is particularly prominent in indoor farms. A few years earlier, I had worked in the agriculture sector as part of a Y Combinator-backed nonprofit, SunFarmer. I combined domain expertise along with professional experience to create a new solution for climate management specific to the indoor farming sector.

What's new about what you're making? What substitutes do people resort to because it doesn't exist yet (or they don't know about it)?

At the moment indoor farm operators cannot easily connect their farm management system with existing building technology. Ambient is the first to provide a seamless way to connect the two. At the moment a lot of the equipment controls are done manually at the site. This is particularly difficult for farm operators that need to frequently adjust environmental conditions based on the plant's unique schedule. Ambient has developed an advanced optimization model using latest technique to overcome this challenge. Our system not only saves on labour costs and energy, but balances these goals with creating optimal conditions for increased plant yield and quality.

Who are your competitors, and who might become competitors? Who do you fear most?

Our biggest competitor are not large building technology providers (e.g., , , Bosch, etc.). Instead, we are competing with other

startups who currently provide indoor farm management technologies, such as lighting systems. We believe Ambient's holistic approach to climate optimization and domain expertise distinguishes us from these competitors.

What do you understand about your business that other companies in it just don't get?

Ambient understands how to help indoor farm operators increase yields and quality through building technology. The role of climate has been overlooked in existing agricultural processes. At the same time, existing smart buildings startups primarily focus on reducing energy costs through their technology. This can often be in direct competition with other indoor climate goals, such as creating optimal indoor climate. Ambient is the first to balance multiple competing optimization goals for indoor farms.

How do or will you make money? How much could you make?

(We realize you can't know precisely, but give your best estimate.)

Vertical farming is a rapidly growing market worldwide with estimated global CAGR of 24%. Asia-Pacific is the global leader in terms of market value with value of about $2.56B. We estimate we can be highly competitive in this market by starting out in Singapore and expending regionally.

At the same time, about 80% of the Ambient's technology is applicable to other building types. This means Ambient is positioned well to expand beyond vertical farms and become a more general smart buildings technology provider – another very large market with potential for disruption.

Ambient's business model consists of CAPEX and OPEX components. Customers pay for the related hardware necessary to enable the system up-front, as a capital cost. Then the software platform and optimization model is delivered through a subscription model. The pricing for the product will be done on a per-area basis, meaning that larger facilities pay more for the service than smaller ones. We are currently refining the exact numbers based on customer conversations, but we can discuss further, along with financial projections, in the follow-on conversations.

How will you get users? If your idea is the type that faces a chicken-and-egg problem in the sense that it won't be attractive to users till it has a lot of users (e.g. a marketplace, a dating site, an ad network),

how will you overcome that?

Our sales model is fairly straightforward at this time. We provide the solution directly to the user based on a standard service model. We started off by signing LOIs with pilot partners. We plan to grow through word-of-mouth referrals and through direct outreach. Since this is an industrial-scale product, some customization of the solution may be needed to acquire new customers. We plan to expand the product features gradually as the expand, meaning that we can generate more value through a long-term synergistic partnership with existing customers.

EQUITY

Have you incorporated, or formed any legal entity (like an LLC) yet?

No

If you have not formed the company yet, describe the planned equity ownership breakdown among the founders, employees and any other proposed stockholders. If there are multiple founders, be sure to give the proposed equity ownership of each founder.

(This question is as much for you as us.)

At the moment we have eight people involved with the startup part-time. Some are directly part of the team while others are in an advisory role. The reason that we have not defined an equity structure yet is because I have been working on the idea mostly on my own for around a year. The team members have started to contribute a few months ago. We are interested in discussing appropriate co-founding arrangement based on contribution so far and planned commitment. Ideally, it would be an even split among the co-founders; however, the breakdown should reflect time and cost invested so far as well as commitment and interest in the future of the company.

Please provide any other relevant information about the structure or formation of the company.

We are participating at a university-run incubator for deep-tech startups. We have not taken investment from them yet, but are planning to pitch in May for $150k.

LEGAL

Are any of the founders covered by noncompetes or intellectual property agreements that overlap with your project? If so, please explain.

No but I have consulted university patent office as well as IP lawyers for the core technology behind the product. The patent documentation is prepared and ready to be filed upon raising the round either from university, YC, or another investor/incubator.

Who writes code, or does other technical work on your product? Was any of it done by a non-founder? Please explain.

I do all of the technical work with support from a part-time designer.

Is there anything else we should know about your company?

(Pending lawsuits, cofounders who have left, etc.)

None at this time

OTHERS

If you had any other ideas you considered applying with, please list them. One may be something we've been waiting for. Often when we fund people it's to do something they list here and not in the main application.

Although this idea is primarily focused on indoor farms, about 80% of the underlying technology is applicable to other kinds of buildings. We received interest from on of the largest healthcare facility management companies in Asia for this type of solution. Unfortunately, we need to grow the startup's capabilities before we can properly cater to a customer of that scale.

Please tell us something surprising or amusing that one of you has discovered.

(The answer need not be related to your project.)

On last week's YC startup school call, I learned from a founder that Mexico is planning to legalize cannabis. Although this segment is not a focus for us, it's an indication that indoor farming will only increase in the future.

CURIOUS

What convinced you to apply to Y Combinator? Did someone encourage you to apply?

Most recently I attended Y Combinator's Outreach Spring Tour Event at my current university (National University of Singapore) hosted by Tamanna and
It was good to connect with the YC team and learn about YC's interest in the region. Having worked for a Y Combinator-backed company in Asia in the past, I saw the advantages of joining the YC community.

How did you hear about Y Combinator?

On-campus events
